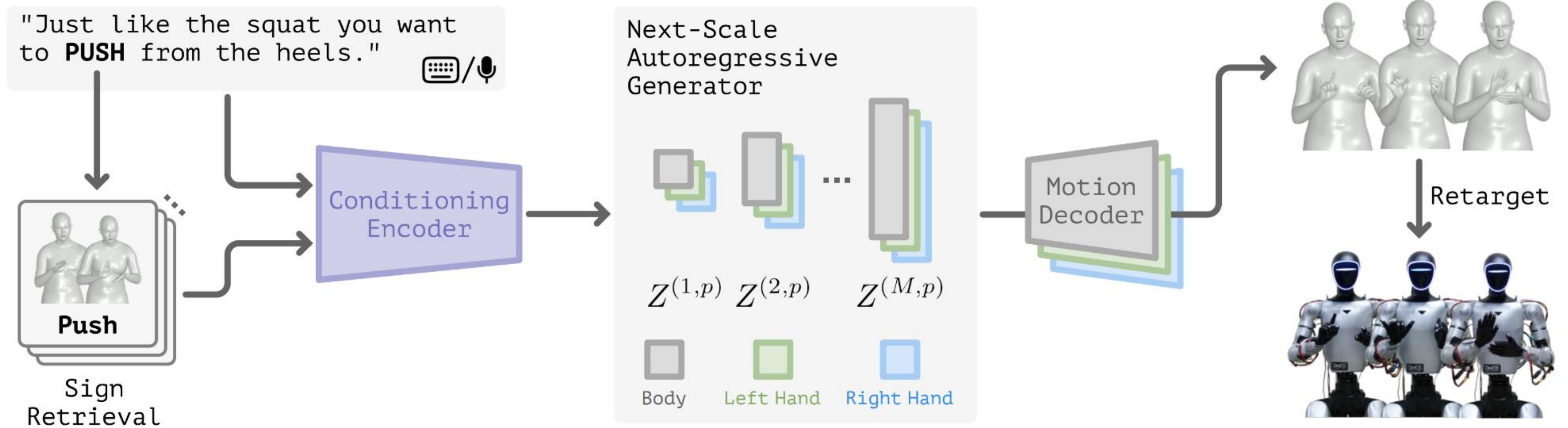


Next-Scale Autoregressive Sign Language Motion Translation for Humanoid Robots

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1. Overview



Novel Task

We introduce Embodied Sign Translation, a new task for translating language into sign motion executable by humanoid robots.

SC + GCC

Self-conditioning and generic context corruption improve robustness to imperfect predictions propagated across scales.

Part-Wise FSQ

Body and bilateral hands are represented with synchronized, part-specific FSQ tokens.

Dynamics Metrics

New metrics characterize motion activity, magnitude, and freeze-tail behavior beyond conventional geometric errors.

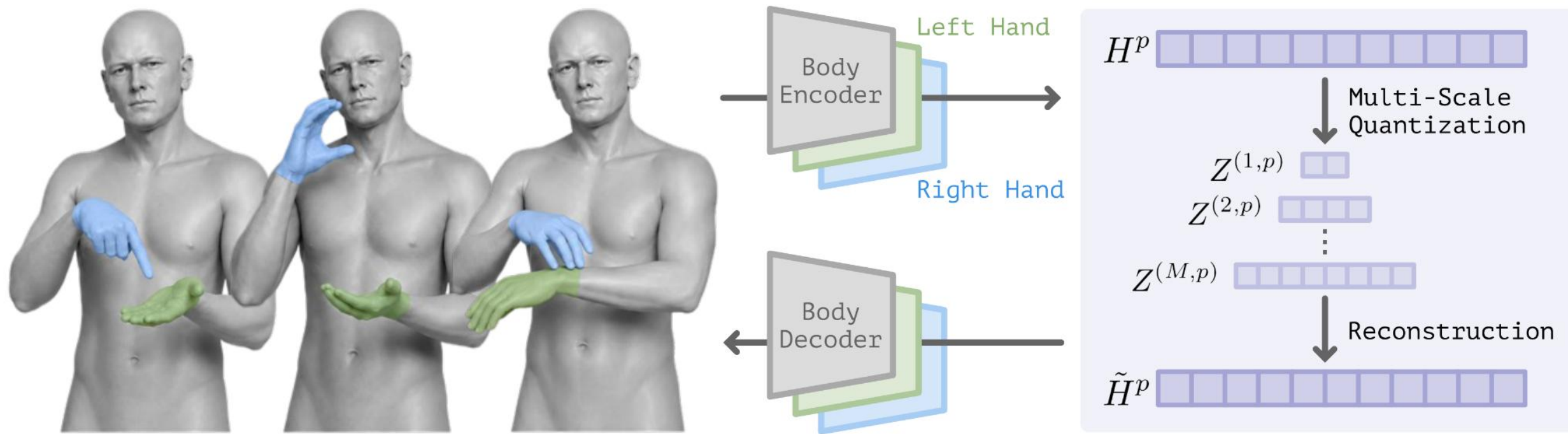
Next-Scale Autoregression

Motion is generated from coarse to fine across temporal scales, progressively refining global structure and fine articulation.

State-of-the-Art Results

Achieve the strongest overall performance across geometric, BT, and dynamics evaluations.

2. Core Method



Part-Wise FSQ

fixed-grid discrete representation for body and hands

Next-Scale AR

coarse-to-fine temporal generation of synchronized streams

SC + GCC

robust conditioning when finer scales depend on imperfect predictions

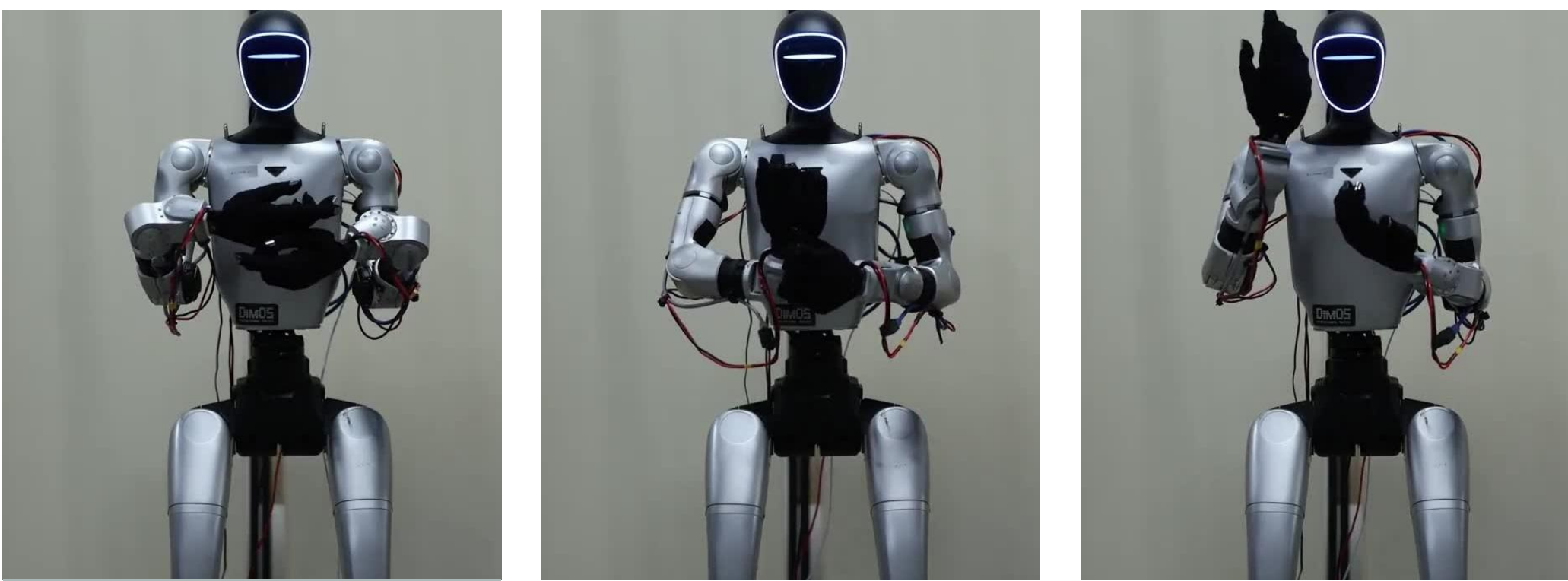
3. Results

Method	DTW-PA-JPE ↓		DTW-JPE ↓		B-T BLEU ↑				Motion Dynamics		
	Body	Hand	Body	Hand	B-1	B-2	B-3	B-4	Dynamic ↔ 0.8	Magnitude ↔ 1	Freeze-tail ↓
GT	–	–	–	–	22.45	8.59	3.95	1.96	0.804	1.000	0.036
PT	10.85	2.66	10.84	10.97	13.44	4.69	1.99	0.91	0.245	0.484	0.232
NAR	4.21	1.79	4.14	5.70	6.81	2.10	0.88	0.42	0.000	0.004	1.000
S-MotionGPT	11.84	3.92	11.91	13.30	12.01	4.33	1.99	1.03	0.005	0.016	0.294
T2S-GPT	4.35	1.82	4.27	5.78	7.13	2.79	1.28	0.60	0.000	0.000	1.000
SOKE	5.73	2.08	5.73	7.26	17.25	6.31	2.86	1.39	0.388	0.515	0.163
Ours	3.96	1.52	3.87	5.34	20.08	7.12	3.04	1.36	0.432	0.523	0.187



4. Real-Robot Execution

Unitree G1 with dual 20-DoF Wuji hands

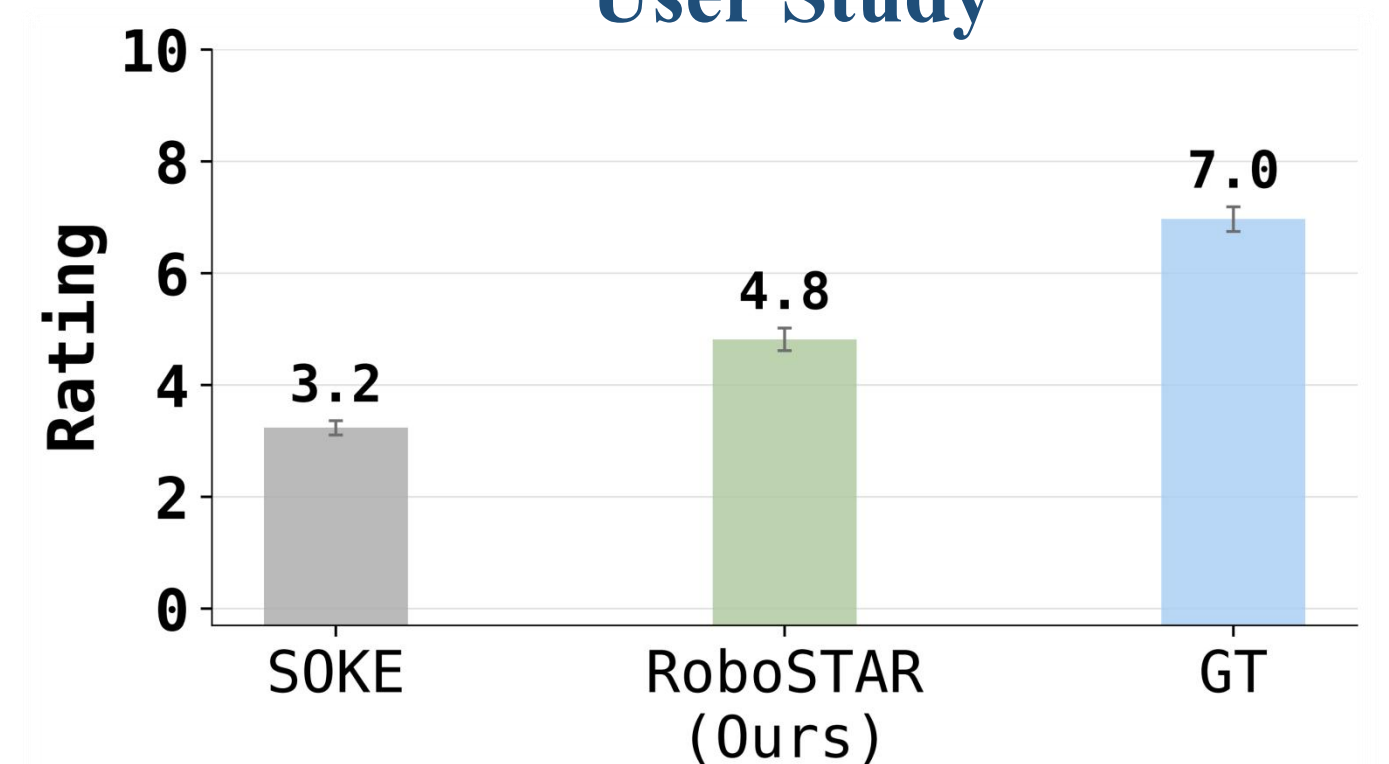


Conclusion

RoboSTAR connects speech/text inputs to coordinated sign-language motion that can be physically executed by humanoid robots.

RoboSTAR achieves state-of-the-art geometric performance on the common How2Sign test subset, while also improving semantic fidelity and motion dynamics.

User Study



Project Page

www.yujiazeng.com/RoboSTAR